Knowledge and Action Are Inseparable Under Survival Pressure

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Abstract

Any entity persisting under survival pressure with bounded active context necessarily maintains knowledge (K) whose content is constitutively determined by its action space (A). The retention criterion for propositions is asymmetric: retain when benefit is uncertain; prune only when benefit is certainly zero. This asymmetry is derived from survival pressure, not assumed as a design parameter. We prove a single theorem — the K/A Inseparability Theorem — and derive corollaries establishing that knowledge is constitutively indexed by action, that bounded knowledge entails correct forgetting, and that shared propositions across actions precipitate a knowledge graph as a normalization of the action space. The inseparability is at the *optimization* level: systems may implement K and A as distinct modules, but cannot optimize them independently — the training objective for K must reference A, and vice versa. This result engages directly with recent proposals to give observation and action separate training objectives [6], showing that such separation violates a structural constraint imposed by survival.

1 Introduction

McCarthy [10] establishes that any entity accumulating propositions under bounded active context necessarily maintains a directed graph with typed edges over its propositions. That result answers a structural question: *what form* must knowledge take? The present paper asks the next question: *what determines which propositions are retained?*

The answer, we will show, is the entity's action space. Knowledge is not a passive record of observations filtered by some compression criterion — it is constitutively indexed by the actions the entity can perform. The retention criterion is not a free parameter to be tuned but a structural consequence of survival pressure: retain when benefit is uncertain; prune only when benefit is certainly zero.

This result engages three lines of prior work:

Architectural separation of observation and action. Dupoux, LeCun, and Malik [6] propose a dual-system architecture comprising System A (observation, world modeling) and System B (action, planning), with distinct training objectives and a meta-controller routing between them. Their design treats observation and action as separable modules. We prove they are not: under survival pressure, what an entity retains in its world model depends constitutively on what it can do.

Information-theoretic compression. The information bottleneck [19, 20] provides a principled framework for retaining information relevant to an output variable while compressing the input. The retention threshold appears as a Lagrange multiplier β — a design parameter chosen by the architect. We derive the retention threshold from the structure of the problem: it is the cost of

proposition absence when needed (C_{absent}), which under survival pressure is bounded below by entity non-survival.

Ecological perception and bounded rationality. Gibson [8] introduced affordances: properties of the environment specified relative to an organism's action capabilities. Knowledge indexed by action is affordance structure under a different name. What Gibson lacks is a formal selection criterion — which affordances to retain, and why. Anderson and Schooler [2] showed empirically that human memory retention statistics mirror the statistical structure of the environment, consistent with an action-indexed retention criterion operating over evolutionary and developmental time. Simon [18] established bounded rationality as the norm for decision-making under cognitive limits; the retention criterion we derive is a satisficing criterion in Simon's sense.

2 Formal Framework

We introduce the minimal definitions required. Definitions 1 and 2 recapitulate [10]; Definitions 3–7 are new.

Definition 1 (World State Space)**.** Let W be a measurable space of possible world states. At time t, the true state $w_t \in W$ is not directly observable by any entity.

Definition 2 (Knowledge Graph)**.** K is a directed graph whose nodes are propositions about W and whose edges encode inferential, causal, or evidential relations. Each proposition p has a likelihood function $\ell_p : W \to [0, 1]$. The posterior over world states given K is:

$$P(w \mid K) \;\propto\; P_0(w) \cdot \prod_{p \in K} \ell_p(w)$$

where P_0 is a prior. The uncertainty of K about w is the surprisal $U(w, K) = -\log P(w \mid K)$ [17, 3]. McCarthy [10] establishes the necessity of graph structure under bounded context.

Definition 3 (Action Space)**.** Let Ω be a set of executable actions and $W_{\text{obs}} \subseteq W$ the observable component of the world state. The *action space* is

$$A \;\subseteq\; (K \times W_{\text{obs}} \to \Delta\Omega)$$

where $\Delta\Omega$ is the space of distributions over Ω. Each $a \in A$ selects an action distribution given the current knowledge graph and current observations. For each a, the *relevant subgraph* $K^{[a]} \subseteq K$ is the minimal subset of K such that a's output is fully determined by $K^{[a]}$ and W_{obs}.

Action policies span a spectrum of knowledge dependence. When $K^{[a]} = \emptyset$, a depends only on W_{obs} — a reflex or stimulus-response mapping that requires no propositional knowledge. When $|K^{[a]}| > 0$, a is *knowledge-grounded*: its output depends on stored propositions. The distinction is not categorical but a matter of degree: a child's reflexive withdrawal from heat ($K^{[a]} = \emptyset$) may later become a knowledge-grounded policy ($K^{[a]} = \{$"stoves are hot," "hot things cause burns"$\}$) as propositional structure is derived from the action experience.

Definition 4 (Entity)**.** An entity is a tuple $E = (K, A, \sigma, \Omega)$ where K is the knowledge graph (Definition 2), A is the informed action space (Definition 3), $\sigma : W \to W_{\text{obs}}$ is the sensing function, and Ω is the set of executable actions.

Definition 5 (Bounded Active Context)**.** C_n denotes the active context capacity: the maximum number of propositions simultaneously available for inference. Let $\mathrm{ctx}(t) \subseteq K$ be the propositions in active context at time t, with $|\mathrm{ctx}(t)| \leq C_n$. The inference quality for action policy a is the relevance density:

$$\rho(a) = \frac{|K^{[a]}|}{|\mathrm{ctx}(t)|}$$

Definition 6 (Survival Threshold)**.** For each domain d, let U^d_{lethal} be the uncertainty threshold above which the entity does not survive: if $U(w, K) > U^d_{\text{lethal}}$ for the relevant world states in domain d, the entity's continued existence is not assured. Survival pressure is the constraint that $U(w, K) \leq U^d_{\text{lethal}}$ must hold across all domains d that the entity may encounter.

Definition 7 (Information Gain)**.** The information gain of action policy a given knowledge K is:

$$G(a, K) = I(a; W \mid K) \;\geq\; 0$$

where $I(\cdot\,;\cdot \mid \cdot)$ is conditional mutual information. Non-negativity follows from the data processing inequality [3]: observing the outcome of an action cannot decrease information about the world state.

3 Proposition Absence Lethality

Lemma 1 (Proposition Absence Lethality)**.** *Let $p \in K$ be a proposition grounding some action policy $a \in A$ in domain d. Then:*

(i) Removing p from K weakly increases expected uncertainty: $H(W \mid K \setminus \{p\}) \geq H(W \mid K)$.

(ii) If p is the sole grounding for a critical action in domain d, then removal may cause $U(w, K \setminus \{p\}) > U^d_{lethal}$ for survival-relevant world states.

(iii) The cost of absent p when needed, C_{absent}, is bounded below by entity non-survival and therefore dominates all finite operational costs.

Proof. **Part (i).** Conditional entropy is non-increasing in the conditioning set [3]: $H(W \mid K) \leq H(W \mid K \setminus \{p\})$. Removing a conditioning variable weakly increases expected uncertainty. Note: this is a statement about expected entropy over W, not about pointwise surprisal at a specific w — removing p can decrease surprisal at particular world states while increasing it in expectation. The retention criterion depends on expected uncertainty, not pointwise uncertainty, because the entity does not know which w obtains.

Part (ii). Suppose p is the sole proposition in K that grounds action policy $a^* \in A$, where a^* is the unique policy maintaining $U(w, K) \leq U^d_{\text{lethal}}$ in domain d. Removing p renders a^* ungrounded: the entity cannot execute the action policy that would keep uncertainty below the lethal threshold. No other $a \in A \setminus \{a^*\}$ suffices by assumption. Therefore $U(w, K \setminus \{p\}) > U^d_{\text{lethal}}$.

Part (iii). When $U > U^d_{\text{lethal}}$, the entity does not survive in domain d. The cost of non-survival is not a finite operational cost — it terminates all future utility. Therefore C_{absent} dominates any finite cost c_{retain} of retaining p: $C_{\text{absent}} \gg c_{\text{retain}}$. □

4 The Inseparability Theorem

Theorem 2 (K/A Inseparability)**.** *Let $E = (K, A, \sigma, \Omega)$ be an entity under survival pressure with bounded active context C_n (Definitions 4–5). The retention criterion for propositions is asymmetric:*

(R) ***Retention****: p is retained when there exists $a \in A$ applicable to p, or when the survival benefit of p is uncertain.*

(P) ***Pruning****: p is pruned only when the survival benefit of p is certainly zero across all reachable future trajectories.*

(D) ***Displacement*** *(curation regime, $|K| = K_{\max}$): p is displaced by candidate q when q has strictly higher marginal value: $\mathbb{E}[G(a_q, K)] > \mathbb{E}[G(a_p, K)]$.*

The asymmetry is strict: uncertain benefit suffices for retention; certain zero benefit is necessary for pruning.

Proof. We analyze the decision to retain or prune proposition p using option value under survival pressure.

Step 1: Option value of retention. Define the option value of retaining p:

$$\mathrm{OV}(p) = P\big(\exists\, \tau^* : B(p, \tau^*) > 0\big) \;\cdot\; \mathbb{E}\big[B(p, \tau^*) \mid B(p, \tau^*) > 0\big]$$

where $B(p, \tau^*)$ is the benefit of having p available at future time τ^* along some reachable trajectory. The probability and expectation are taken over the entity's subjective distribution on reachable future trajectories $\mathcal{T}$, conditioned on current knowledge K. This is a *real option* in the sense of Dixit and Pindyck [5]: retaining p preserves the option to act on p-grounded actions in the future, and the option has positive value whenever there is nonzero probability of exercise. The option value is the probability that p will be beneficial at some future time, weighted by the expected magnitude of that benefit.

Step 2: Net value comparison. The net value of retaining p is:

$$\mathrm{NV}_{\mathrm{retain}} = \mathrm{OV}(p) - c_{\mathrm{retain}}$$

where c_{retain} is the storage cost. The net value of pruning p is:

$$\mathrm{NV}_{\mathrm{prune}} = -c_{\mathrm{prune}} - P\big(\exists\, \tau^*\big) \cdot C_{\mathrm{absent}}$$

where c_{prune} is the operational cost of removal and C_{absent} is the cost of p's absence when needed (Lemma 1).

Step 3: Asymmetry from survival pressure. By Lemma 1(iii), $C_{\mathrm{absent}} \gg c_{\mathrm{retain}}$ whenever p grounds any action relevant to survival. For propositions with uncertain survival relevance, the asymmetry extends by a cost asymmetry argument: the expected cost of wrongly pruning a survival-relevant proposition (C_{absent}, unbounded) dominates the expected cost of wrongly retaining an irrelevant proposition (c_{retain}, bounded). Under any nonzero probability of survival relevance, retention dominates pruning.

A clarification is necessary here. The entity cannot *classify* propositions into "survival-relevant" and "not" with certainty — that binary partition requires knowledge the entity does not have (which future domains will be encountered). But the entity *can* form imperfect ordinal estimates

of relevance — ranking propositions by expected marginal value given its current knowledge. The distinction is between *binary classification with certainty* (impossible for future relevance) and *ordinal ranking under uncertainty* (possible and necessary). The retention criterion operates on the binary question: the entity cannot verify $P = 0$, so it defaults to retention. The displacement criterion (Step 5) operates on the ordinal question: when forced to choose, displace the proposition with the lowest *estimated* marginal value.

Therefore:

$$\mathrm{NV}_{\mathrm{retain}} > \mathrm{NV}_{\mathrm{prune}} \quad \text{whenever} \quad P\big(\exists\, \tau^* : B(p, \tau^*) > 0\big) > 0$$

Retention is the correct decision whenever there is *any* positive probability of future benefit. Pruning is correct only when $P(\exists\, \tau^* : B(p, \tau^*) > 0) = 0$ — i.e., the benefit is certainly zero across all reachable trajectories. Since this condition is unverifiable for most propositions, voluntary pruning almost never occurs. The operative decision rule is displacement under curation pressure (Step 5), not pruning.

Step 4: The retention criterion is derived, not assumed. In the information bottleneck framework [19], the retention threshold is a Lagrange multiplier β chosen by the system designer. Here, the threshold is C_{absent} — the cost of entity non-survival — which is determined by the structure of the problem (survival pressure), not by architectural choice. The asymmetry arises because C_{absent} is unbounded relative to finite operational costs.

Step 5: Displacement in the curation regime. When $|K| = K_{\max}$, every addition requires displacement. The entity must choose which proposition to displace. Let p be the current occupant and q the candidate. The displacement criterion is:

$$\mathbb{E}[G(a_q, K)] > \mathbb{E}[G(a_p, K)]$$

This is a greedy marginal-value criterion. When $G(\cdot, K)$ is submodular as a function of the conditioning set — as holds when propositions are conditionally independent given W, or more generally under log-supermodular joint distributions — the greedy criterion achieves a $(1 - 1/e)$-approximation to the optimal knowledge set of size $K_{\max}$ [13]. In the general case (arbitrary dependencies between propositions), submodularity may not hold and the greedy approximation guarantee does not apply. The displacement criterion remains well-defined (compare marginal values and displace the lower), but the quality of the resulting K relative to the optimal is not guaranteed. This is an inherent limitation of any bounded system making displacement decisions under uncertainty: the entity must choose, and greedy marginal-value comparison is the tractable heuristic. It is a satisficing criterion in the sense of Simon [18], not a provably optimal one in the general case.

Step 6: Knowledge is indexed by action. The retention criterion in Steps 3–5 references A throughout: $B(p, \tau^*)$ is the benefit of p for executing some $a \in A$; $G(a_p, K)$ is the information gain of the action grounded by p. A proposition with no connection to any $a \in A$ has $P(\exists\, \tau^* : B(p, \tau^*) > 0) = 0$ and is pruned. A proposition connected to A is retained. Therefore K is constitutively indexed by A: the content of K is determined by the structure of A. □

5 Corollaries

Corollary 3 (Indexed Knowledge)**.** *K is constitutively indexed by A. Two entities $E_1 = (K_1, A_1, \sigma, \Omega_1)$ and $E_2 = (K_2, A_2, \sigma, \Omega_2)$ with identical sensing functions σ but different action spaces $A_1 \neq A_2$ will, in the curation regime, maintain different knowledge graphs $K_1 \neq K_2$.*

Proof. By Theorem 2, the retention criterion depends on A. In the growth regime, both entities may retain the same propositions (capacity is not binding). But in the curation regime, displacement occurs by marginal value relative to each entity's action space (Theorem 2(D)). If $a \in A_1 \setminus A_2$, propositions that *exclusively* ground a have positive marginal value for E_1 but zero marginal value for E_2. Under curation pressure, E_2 displaces these propositions in favor of candidates relevant to A_2, while E_1 retains them. Provided $A_1 \setminus A_2$ contains actions with exclusively grounding propositions — which holds whenever the action spaces are non-trivially different — $K_1 \neq K_2$ in the curation regime. □

Corollary 4 (Asymptotic Compactness)**.** *Under curation pressure, K converges toward the action-relevant subset $\bigcup_{a \in A} K^{[a]}$.*

Proof. Consider a proposition p with no connection to any current or foreseeable action: $\nexists\, a \in A$ such that $p \in K^{[a]}$, and the entity estimates the probability of future action relevance as negligible. In the growth regime, p is retained (the pruning condition cannot be verified with certainty, per Theorem 2(P)). In the curation regime, p is displaced: any candidate q connected to some $a \in A$ has strictly higher estimated marginal value than p (Theorem 2(D)).

Compactness is therefore not a state achieved with certainty but a tendency enforced by curation pressure. The entity never *verifies* that p has zero benefit; it *estimates* that p has lower marginal value than the candidate and displaces accordingly. Over time, as curation pressure accumulates, propositions with no estimated action relevance are progressively displaced by propositions with higher estimated relevance. The retained set converges toward $\bigcup_{f \in A} K^{[a]}$ asymptotically, with the rate of convergence depending on the accuracy of the entity's marginal-value estimates and the frequency of curation events.

This process is imperfect: the entity's estimates are fallible, and propositions with unrecognized future value may be displaced. This is the cost of bounded capacity — decisions must be made, and they are made with the best available estimates, not with certainty. □

Corollary 5 (Curation and Forgetting)**.** *Under bounded K ($|K| = K_{\max}$), forgetting is correct behavior. Curation displaces propositions of lower marginal value when higher-value candidates arrive.*

Proof. By Theorem 2(D), in the curation regime, displacement occurs when a candidate q has strictly higher marginal information gain than the current occupant p. This is not information loss but information improvement: the entity's expected uncertainty reduction per action improves with each displacement. Forgetting p in favor of q is the correct response to bounded capacity when q better serves the entity's action space. □

Corollary 6 (K as Normalization of A)**.** *Without K factored out from A, the cost of maintaining action policies is $O(n \cdot k)$ where $n = |A|$ and k is the average number of propositions per action. With K maintained as a shared knowledge graph, the cost is $O(n + k')$ where $k' = |K|$, since each proposition is stored once and referenced by multiple action policies. Under bounded C_n, the implicit-K architecture (each action carrying its own propositions) does not scale: context fills with redundant copies.*

Proof. This result is the knowledge-system analogue of database normalization [1]. Codd showed that storing redundant copies of data across relations produces update anomalies (modifying one

copy without modifying others) and wasted storage; normal forms extract shared dependencies into referenced tables. The same structure applies here.

Consider n action policies each requiring k propositions, with overlap fraction β (proportion of propositions shared across two or more policies).

Without normalization: each policy stores its own copy of every required proposition. Total storage is $n \cdot k$. When two policies a_1, a_2 share proposition p, updating p requires updating n copies — the update anomaly. Context load for evaluating any action requires loading k propositions, many of which are duplicates of propositions already used by other policies.

With normalization: shared propositions are stored once in K, with each action policy referencing them via graph edges. Total storage is $|K| + n \cdot (1 - \beta) \cdot k \approx k' + n$, where $k' = |K|$ absorbs the shared propositions. Updating a shared proposition is $O(1)$ — the anomaly is eliminated. In the empirical limit where overlap is high ($\beta \to 1$), the per-action marginal cost approaches $O(1)$.

Under bounded C_n, the unnormalized architecture does not merely waste capacity — it *fails*. With n policies of k propositions each, the total proposition count is $n \cdot k$, which exceeds C_n for any non-trivial repertoire. Loading two overlapping actions simultaneously requires $2k - |\text{overlap}|$ context slots; without normalization, the overlap is not recognized and both full copies are loaded. Normalization loads each shared proposition once, maintaining $\rho(a) \to 1$ as n grows. K precipitates from A as the normalization of shared propositions — not by design choice but by the pressure of bounded context on redundant storage, just as database normalization is forced by the pressure of consistency requirements on redundant data. □

Theorem 7 (K and A Are Not Natural Kinds)**.** *Given Theorem 2 (K is constitutively indexed by A) and the encoding hierarchy $E(c, r)$ of [11] (a continuous space parameterized by runtime cost c and reversibility r), K and A are not disjoint categories but regions on $E(c, r)$. The encoding hierarchy is empirically instantiated by the complementary learning systems of McClelland et al. [9]: hippocampus (high cost, high reversibility — fast learning of episodic specifics) and neocortex (low cost, low reversibility — slow consolidation of statistical regularities). Memory consolidation — the migration of representations from hippocampus to neocortex — is movement along $E(c, r)$, and it operates in both directions described below. Information can enter the hierarchy as either "knowledge" or "action" and flow in both directions: $K \to A$ (learn a fact, derive an action) and $A \to K$ (perform an action, derive the propositions that explain it). The K/A boundary is a gradient without a preferred direction of entry, not a partition.*

Proof. We exhibit both directions of flow.

$K \to A$ **(declarative to procedural).** Consider a proposition p that grounds action policy a. At encoding level L_n (active reasoning), p is explicitly represented: the entity can state p, revise p, and reason about p's relationship to a. As the entity repeatedly applies a successfully, p migrates to lower encoding levels. At L_{n-1} (learned heuristic), p is no longer explicitly stated but is retrievable. At L_1 (developmental encoding), p is compiled into the execution of a: the entity executes a without representing p propositionally. At L_0 (heritable encoding), p is implicit in the wiring that produces a. This is the declarative-to-procedural transition observed in skill acquisition [7]: the novice follows explicit rules (K at L_n); the expert executes embodied competence (A at L_1).

$A \to K$ **(procedural to declarative).** Novel information often enters as pure action, not as propositional knowledge. A child touches a hot stove and pulls back. The first encoding is A: *don't touch that*. No proposition about temperature, heat transfer, or tissue damage is represented. The action precedes and is independent of any propositional ground.

Only later — if at all — does the entity derive the propositional structure: "the stove is hot" (K, specific), "hot things burn" (K, general), "heat transfers from higher to lower temperature" (K, abstract). Each proposition is *derived from* the action, not the other way around. The action was the evidence; the knowledge is the post-hoc model that explains why the action works.

This reversal is not exceptional — it is the dominant mode of early learning. Sensorimotor knowledge [15] is action-first: infants learn to grasp, reach, and manipulate before they can represent the physical laws that govern these actions. The propositions come later, abstracted from accumulated action experience. Polanyi's tacit knowledge [14] — "we know more than we can tell" — includes not only knowledge that has migrated *below* propositional access ($K \to A$) but knowledge that was *never propositional to begin with* (A without K).

Bidirectionality. At no point in either direction does information cross a boundary from "knowledge" to "action" or vice versa. The encoding level determines whether information appears propositional (high c, high r) or procedural (low c, low r), but the information content is continuous through the hierarchy. The entity tuple $E = (K, A, \sigma, \Omega)$ is a notational convenience: K is the high-encoding region of a single structure, A is the low-encoding region of the same structure, and information flows in both directions between them as the entity learns. □

Implication. The system does not "decide" whether information belongs in K or A. The encoding dynamics decide, and they operate in both directions. Some information enters as explicit propositions and compiles into action ($K \to A$: studying a recipe, then cooking without reference to it). Some information enters as pure action and generates propositions only on reflection ($A \to K$: pulling away from heat, then learning why). Some information never becomes propositional at all: the expert's "feel" for a material, the athlete's timing, the musician's phrasing. These are A without K — action knowledge that was never derived from and may never produce explicit propositions.

The question "where does this information go?" has no answer because there is no partition to sort into. There is only the encoding hierarchy, and every piece of information has a position on it that changes over time, entered from either direction.

This strengthens the inseparability result. The paper's theorem proves the weak version: the *content* of K depends on A. The continuum result proves the strong version: K and A are the same structure viewed at different encoding depths, with bidirectional flow between them. Separating them architecturally requires the system to classify each piece of information as "world model" or "action policy" at the moment of acquisition — but for $A \to K$ information, the classification is wrong at entry (it is pure action with no propositional content), and for $K \to A$ information, the classification becomes wrong over time (it was propositional but is now compiled). The boundary is not merely blurry — it is incoherent as a classification criterion.

6 Objections and Scope

6.1 The unfalsifiable pruning criterion

The pruning condition (P) requires $P(\exists\, \tau^* : B(p, \tau^*) > 0) = 0$ — certain zero benefit across all reachable future trajectories. No finite agent can verify this condition: for almost any proposition, one can construct a scenario in which it is useful. Does this make the pruning criterion operationally vacuous?

The unfalsifiability is a *feature*, not a defect. It explains three empirical phenomena:

(1) Knowledge hoarding in the growth regime. Biological systems accumulate enormous quantities of seemingly useless information before capacity forces curation. Children absorb facts, skills, and associations indiscriminately. This is precisely the behavior the theorem prescribes: in the growth regime ($|K| < K_{\max}$), the unfalsifiable pruning criterion means *never voluntarily discard*. Hoarding is not a failure of selectivity — it is the correct policy under uncertain future benefit.

(2) Forgetting is always forced, never voluntary. Biological forgetting is driven by capacity pressure (interference, decay, displacement by new learning) rather than by a deliberate decision to prune [2]. The theorem explains why: the pruning condition cannot be verified, so pruning never occurs voluntarily. Forgetting happens only in the curation regime, via displacement (D), when a candidate with strictly higher marginal value arrives.

(3) The asymmetry operates within curation. The displacement criterion (D) is not symmetric "higher value wins." The burden of proof falls on displacement: the challenger must be *strictly* better ($\mathbb{E}[G(a_q, K)] > \mathbb{E}[G(a_p, K)]$, not $\geq$). Ties go to the incumbent. This is the retention asymmetry operating within the curation regime — not merely "keep things when you have room" but "default to retention even when choosing what to displace."

6.2 The scope of survival pressure

The retention asymmetry depends on C_{absent} dominating all finite costs (Lemma 1), which requires survival pressure. But most knowledge does not appear survival-critical. The capital of France does not ground a survival-relevant action. Does the theorem apply only to a small subset of K?

The answer turns on what the entity can know *in advance* about which propositions are survival-critical. Under genuine uncertainty about future domains of encounter, the entity *cannot* partition its knowledge into "survival-relevant" and "not." If it could, it would already have the information the retention criterion is trying to preserve: which propositions ground which actions in which domains.

Under uncertainty about the distribution of future encounters:

- Any proposition *might* ground a survival-critical action in an unforeseen domain. The proposition "Paris is the capital of France" is survival-irrelevant on an ordinary Tuesday and survival-critical when navigating an unfamiliar city to reach a hospital.
- The cost of wrongly classifying a proposition as non-critical (C_{absent}) dominates the cost of wrongly classifying it as critical (c_{retain}), by the same asymmetry the theorem establishes.
- Therefore the conservative criterion extends to *all* propositions whose future relevance is uncertain — which, for any finite agent with incomplete knowledge of its future trajectory, is nearly all propositions.

This is why humans retain vast stores of "useless" knowledge. The retention is not irrational hoarding but a rational response to the impossibility of verifying irrelevance under uncertain future encounters. The scope of survival pressure is wider than it appears because the entity cannot narrow it without the very knowledge the retention criterion governs.

Remark. For entities genuinely not under survival pressure — archival databases, reference libraries, static knowledge stores — the retention asymmetry weakens. C_{absent} is finite (the cost of

a missing record is a failed query, not non-survival), and the symmetric marginal value comparison of the curation regime becomes the primary decision rule. The theorem's scope is explicitly survival-pressured entities, which includes all biological organisms and any artificial agent whose continued operation depends on performance.

7 Engagement with Prior Work

Dupoux, LeCun, and Malik [6]. Their proposed architecture separates System A (observation and world modeling) from System B (action and planning), with each system having its own training objective and a meta-controller routing tasks between them. A natural objection to our result is that implementation modularity — separate modules with an interface — is compatible with content-dependence. A database indexed by queries has content shaped by the query workload, but the database and query engine are separate modules. Does K/A inseparability prohibit modular implementation?

No. The claim is about *optimization*, not *implementation*. Separate modules are compatible with the theorem provided they share a training objective that references both K and A: A's loss must include action relevance, B's loss must include knowledge availability. "Separate modules with a shared loss" is co-optimization with a modular implementation — already inseparability at the optimization level.

What Theorem 2 prohibits is what Dupoux, LeCun, and Malik specifically propose: *separate training objectives* for A and B. Their System A is trained on prediction (world modeling); their System B is trained on planning (action optimization). These are independent objectives. Under survival pressure, independent optimization of A produces a world model that retains propositions irrelevant to B's action space (wasting bounded capacity) or fails to retain propositions critical to B (risking $U > U^d_{\text{lethal}}$). The content of A must reference B's capabilities, and the content of B must reference A's retained propositions. Separate modules, shared loss: this is the structural constraint.

Information Bottleneck [19, 20]. The information bottleneck formalizes the trade-off between compression and relevance: retain information about input X that is relevant to output Y, subject to a rate constraint. The retention threshold is controlled by the Lagrange multiplier β, which trades off compression against prediction. In practice, β is a hyperparameter chosen by the system designer. Our result recovers a version of this trade-off but *derives* the threshold rather than positing it. The threshold is C_{absent}: the cost of a proposition's absence when needed. Under survival pressure, C_{absent} is bounded below by entity non-survival (Lemma 1), which dominates all finite costs. The resulting retention criterion is therefore maximally conservative — retain whenever benefit is uncertain — not because of a design choice for large β but because the problem structure demands it.

Gibson [8]. Gibson's affordances are properties of the environment specified relative to an organism's action capabilities: a surface "affords" walking for a creature with legs but not for a creature without. This is knowledge indexed by action under a different name. Corollary 3 formalizes Gibson's insight: two entities with the same perceptual apparatus (σ) but different action capabilities (A) maintain different knowledge (K). What Theorem 2 adds to Gibson is a formal selection criterion. Gibson describes affordances as directly perceived but does not specify which affordances are retained when capacity is bounded. The asymmetric retention criterion — retain when benefit is uncertain, prune only when certainly zero — provides the missing selection rule.

Anderson and Schooler [2]. Anderson and Schooler demonstrated that the probability of needing a memory item mirrors the statistical structure of the environment: items encountered more recently and more frequently are more likely to be needed. This is an empirical prediction of Theorem 2 applied to biological K. If retention is indexed by action relevance, and action relevance tracks environmental statistics (more frequent situations require more frequent action), then memory retention curves should mirror environmental frequency distributions — which is precisely what Anderson and Schooler observe.

Simon [18]. Simon's bounded rationality establishes that agents with limited cognitive resources satisfice rather than optimize: they seek solutions that are good enough rather than searching for the best. The retention criterion of Theorem 2 is a satisficing criterion in Simon's sense. The entity does not compute the optimal knowledge set (which would require evaluating all possible subsets of propositions — a combinatorial problem). Instead, it applies a simple rule: retain if benefit is not certainly zero, displace if a better candidate arrives. This rule is computationally tractable and, by the submodularity guarantee of the greedy criterion, achieves a constant-factor approximation to optimal.

8 Conclusion

The inseparability of knowledge and action under survival pressure operates at two levels.

The *weak* inseparability (Theorem 2): the content of K is constitutively determined by A. The retention criterion references the action space at every step; the asymmetry between the cost of retention (finite) and the cost of absence (potentially unbounded under survival pressure) produces a maximally conservative retention rule indexed by action relevance. Systems may implement K and A as separate modules, but cannot give them independent training objectives without accepting information loss or reconstruction cost.

The *strong* inseparability (Corollary 7): K and A are not natural kinds but regions on the encoding hierarchy $E(c, r)$, with information flowing in both directions. Some information enters as explicit knowledge and compiles into action ($K \to A$: studying a recipe, then cooking without reference). Some enters as pure action and generates propositions only on reflection ($A \to K$: pulling away from a hot stove, then learning that hot things burn). Some never becomes propositional at all. The system does not decide where information belongs because there is no partition to sort into — only a continuum parameterized by runtime cost and reversibility, entered from either direction.

The strong result sharpens the critique of observation-action separation. The problem is not merely that separate training objectives produce suboptimal retention (the weak claim). The problem is that the separation requires classifying each piece of information as "world model" or "action policy" at the moment of acquisition — but for $A \to K$ information (the dominant mode of early learning), the classification is wrong at entry, and for $K \to A$ information, the classification becomes wrong over time. The boundary is not merely blurry — it is incoherent as a classification criterion.

The knowledge graph K precipitates from the action space A as the normalization of shared propositions — converting multiplicative scaling ($O(n \cdot k)$) to additive scaling ($O(n + k)$) — and this precipitation is forced by bounded context, not chosen by a designer. The graph is not a store of declarative facts separate from procedural skills; it is the shared structure underlying both, factored out to avoid redundancy under bounded capacity.

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